



Redacted

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Abstract

We propose a Redacted — a domain-specific conversational AI that recommends PC components based on user needs and budget. General-purpose LLMs often provide inaccurate component advice due to stale or hallucinated specifications - we will build an assistant with explicit compatibility rules to ground recommendations in verified manufacturer data.

Keywords

PC, hardware, computers, domain specific AI, assistant

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Introduction

Choosing PC components is a common technical challenge for newcomers to the hobby, yet general-purpose LLMs often provide incorrect information (e.g., CPU socket compatibility, number of PCIe slots, bandwidth...). Our goal is to design an AI assistant that provides trustworthy, source-grounded component recommendations for PC builders. We aim to speed up the component selection process while recommending components that would suit the consumer's needs best.

Proposed Dataset

We will compile a corpus of official product specification pages and manuals from major component manufacturers (ASUS, MSI, Gigabyte, ASRock, AMD, Intel, NVIDIA). For each component we will extract both structured attributes and unstructured text segments (datasheets, compatibility notes, BIOS guidance) to support RAG. The initial scope targets consumer-grade parts released in roughly the last three years so the assistant remains relevant for typical PC builders.

Initial approach

We will scrape and normalize official specification pages into a structured schema plus text chunks, then enforce explicit compatibility rules (socket, chipset, RAM type, PCIe lanes, PSU headroom) to ensure each candidate build is compatible before recommendations are generated. A Retrieval-Augmented Generation layer retrieves the most relevant spec passages for each query, and an instruction-tuned LLM combines that

evidence with rule verdicts to produce conversational recommendations with source citations.

Related work

RAG-based systems demonstrate that injecting retrieved passages before decoding substantially improves factual accuracy on knowledge-intensive tasks, yet still struggles when key specifications are missing [1, 2]. Instruction-tuned models offer better conversational control but inherit the same grounding limitations unless connected to authoritative data sources [3]. Bengio argues for hybrid neural-symbolic approaches to ensure dependable reasoning in high-stakes settings, motivating our combination of RAG with explicit compatibility rules [4]. Finally, tools like the PCPartPicker compatibility engine show that deterministic rule bases prevent hardware mismatches but lack conversational flexibility; our assistant aims to merge their reliability with LLM-driven dialogue [5].

References

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